ESWCap combined C+R front end (STM32G474 carrier)

REV C: USB-only; 4-wire C/R; OPAMP3 SH buffer; J5 FH/SH/SL/FL; PA0=DRIVE, PA6=BIAS_MON

C: DRIVE -> C mux -> Rsense -> Vcap -> C1/C2 -> K1B -> N_DUT -> DUT -> B -> K1A -> GND

REV B: C1 = 1000 uF / 35 V BIPOLAR (+ marking not used).

D6 (1N4148) blocks Q1 body diode; R9 = 47R 2W; C3 returns to VREF.

Console J7: PA9=TX, PA10=RX, GND, +3V3. See docs/pcb-design.md.

R: +3V3 -> Q1 -> N_DUT -> DUT -> B -> K2A -> R mux -> RSENSE -> GND. OPAMP1: VINP=B(PA7), VINM=VREF(PA3), VOUT=PA2. TIA: LMC6482 A.

J2
1 GND
2 +3V3
3
4
5
6 DRIVE
7 VCAP_ADC
8 OP1_OUT
9 VREF
10 BIAS_SET
11 VBIAS
12 TIA_OUT
13 FL
14 C_MODE_SW
15 VN_SENSE
16
17 MUXR_EN
18
19 K2_DRV
20 SSR13
WeAct P2

J1
1 GND
2 K1_DRV
3 MUXR_A1
4 MUXR_A0
5 MUXC_EN
6 MUXC_A1
7 MUXC_A0
8 BIAS_UART_RX
9 BIAS_UART_TX
10 BIAS_EN
11
12
13 UART_RX
14 UART_TX
15 BIAS_MON
16
17
18 R_FORCE
19 SSR2_CTRL
20 VBUF
WeAct P1

U2 74LVC1G34 +3V3 DRIVE 2 4 C_MUX_COM 1 3 GND

U3 MAX4619
C_CH3 1 Y1 VCC 16 +3V3
C_CH2 2 Y0 Y 15 MUXC_Q
MUXC_Q 3 Z1 X 14 MUXC_P
C_MUX_COM 4 Z X1 13 C_CH1
MUXC_P 5 Z0 X0 12 C_CH0
MUXC_EN 6 ENABLE A 11 MUXC_A0
7 NC B 10 MUXC_A0
GND 8 GND C 9 MUXC_A1

R1 C_CH2 1 2 VCAP 98.9 1%
R2 C_CH1 1 2 VCAP 993.6 1%
R3 C_CH0 1 2 VCAP 98.6k 1%
R4 C_CH3 1 2 VCAP 1.01M 1%

C1 VCAP 1 2 C_BLOCK 1000uF/35V BP
C2 VCAP 1 2 C_BLOCK 68nF

D1 BAT54S 1 GND A1 K2 2 +3V3 COM 3 VCAP

J5 1 FH 2 SH 3 SL 4 FL DUT
J6 1 V_BIAS 2 GND V_BIAS 0-20V

R17 +3V3 1 2 VREF 7.15k
R18 VREF 1 2 GND 1k
C5 VREF 1 2 GND 100nF
R19 +3V3 1 2 VBIAS 10k
R21 VBIAS 1 2 GND 10k
C6 VBIAS 1 2 GND 100nF
C8 +3V3 1 2 GND 100nF
C9 +3V3 1 2 GND 10uF
C10 +3V3 2 GND 100nF

R38 1 SH 2 SH_DIV 10M 0.1%
R39 1 SH_DIV 2 GND 1M
C20 1 SH_DIV 2 GND 100pF
C21 1 SH 2 SH_DIV 10pF

FLG1 1 +3V3
FLG3 1 GND
FLG4 1 V_BIAS

U5 AQY212
SSR1_LED 1 LED+ SW 3 SSR1_SW
GND 2 LED- SW 4 V_BIAS

U6 AQY212
SSR3_LED 1 LED+ SW 3 SSR3_SW
GND 2 LED- SW 4 VCAP

U7 AQY212
SSR2_LED 1 LED+ SW 3 SSR2_SW
GND 2 LED- SW 4 FH

R22 SSR13 1 2 SSR1_LED 330
R23 SSR13 1 2 SSR3_LED 330
R24 SSR2_CTRL 1 2 SSR2_LED 330

R9 SSR1_SW 1 2 FH 47
R11 SSR3_SW 1 2 GND 1
R10 SSR2_SW 1 2 GND 100

R12 V_BIAS 1 2 FH 4.8M
R13 +3V3 1 2 Q1_GATE 100k
R14 OP1_OUT 1 2 Q1_GATE 1k
R15 R_FORCE 1 2 Q1_GATE 10k

Q1 BSS84 +3V3 S G D Q1_D
U9 AO3401 +3V3 LED+ SW 3 SH
Q1_D LED- SW 4 VN_IN
U9_LED_K OP1_OUT 1 2 VREF 100pF
AQY212

U10 AQY212
+3V3 LED+ SW 3 SH_DIV
U10_LED_K LED- SW 4 VN_SENSE

D2 BAV199 GND 1 A1 K2 2 +3V3 COM 3 VN_SENSE
R43 1 C_MODE_SW U10_LED_K 330

R20 VN_IN 1 2 VN_SENSE 100k
C3 OP1_OUT 1 2 VREF 100pF
VN_SENSE 1 C7 2 GND 1nF

R41 1 K1_C 2 U9_LED_K 330

U1A LMC6482
VBIAS 3 IN+ OUT 1 TIA_OUT
FL 2 IN- OUT

U1C LMC6482
+3V3 8 V+ V- 4 GND

R26 1 K2_DRV 2 K2_BASE 1k
R28 1 K2_BASE 2 GND 10k
1 SL 2 SL_BUF 100k
U1B LMC6482
SL_BUF 5 IN+ OUT 7 VBUF
VBUF 6 IN-

D4 K1_C 2 1 +3V3 1N4148
R25 K1_DRV 1 2 K1_BASE 1k
R27 K1_BASE 1 2 GND 10k

Q2 BC847 K1_C C B E K1_BASE GND

K1 COIL+ NC1 3 GND
+3V3 COM1 4 FL
NO1 5 C_BLOCK
NC2 10 FH
COM2 9
K1_C 12 COIL- NO2 8

R16 FL 1 2 RF_TOP 10M 0.1%
D3 BAT54S GND 1 A1 K2 2 +3V3 COM 3 TIA_OUT
C4 FL 1 2 RF_TOP 10pF

J8
Pin_1 1 BIAS_SET_F
Pin_2 2 BIAS_EN
Pin_3 3 BIAS_MON_F
Pin_4 4 GND
Pin_5 5 +3V3
Pin_6 6 BIAS_UART_TX
Pin_7 7 BIAS_UART_RX
Pin_8 8 GND

D5 K2_C 2 1 +3V3 1N4148

R34 1 BIAS_SET 2 BIAS_SET_F
C16 1 BIAS_SET_F 2 GND 1k
R36 1 BIAS_MON_F 2 BIAS_MON 10uF
C17 1 BIAS_MON 2 GND 100nF

Q3 BC847 K2_C C B E K2_BASE GND

R35 2 GND 10k 1 BIAS_EN

K2 COIL+ NC1 3
+3V3 COM1 4 FL
NO1 5 MUXR_COM
NC2 10 RF_TOP
COM2 9 TIA_OUT
K2_C 12 COIL- NO2 8
KELVIN

R5 R_CH0 1 2 GND 98.9 0.1%
R6 R_CH1 1 2 GND 993.6 0.1%
R7 R_CH2 1 2 GND 98.6k 0.1%
R8 R_CH3 1 2 GND 4.7M 0.1%

U4 MAX4619
R_CH3 1 Y1 VCC 16 +3V3
R_CH2 2 Y0 Y 15 MUXR_Q
MUXR_Q 3 Z1 X 14 MUXR_P
MUXR_COM 4 Z X1 13 R_CH1
MUXR_P 5 Z0 X0 12 R_CH0
MUXR_EN 6 ENABLE A 11 MUXR_A0
7 NC B 10 MUXR_A0
GND 8 GND C 9 MUXR_A1

J4 1 GND 2 GND 3 +3V3 4 +3V3 WeAct P4 3V3
J3 1 GND 2 GND 3 4 WeAct P3 (NC)

C11 +3V3 1 2 GND 100nF
C12 +3V3 1 2 GND 100nF
C13 +3V3 1 2 GND 100nF
C14 +3V3 1 2 GND 100nF
C15 +3V3 2 GND 10uF

R29 DRIVE 1 2 GND 10k
R30 C_MODE 1 2 +3V3 100k
R31 MUXR_EN 1 2 +3V3 100k
R32 VCAP 1 2 VCAP_ADC 1k

R33 VCAP 1 2 C_BLOCK 1M
D6 Q1_D 2 1 FH 1N4148

J7 1 UART_TX 2 UART_RX 3 GND 4 +3V3 Console

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